

Electrochemical characteristics of B₄C or BN added MnO₂ cathode material for alkaline batteries

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ABSTRACT

Boron compounds generally produce a battery with high energy density. Since boron is an excellent conductor of electricity a battery with a high power density can also be achieved. With this objective, the electrochemical characteristics of boron carbide (B₄C) added manganese dioxide (MnO₂) for use as a cathode in alkaline battery were investigated during the discharge–charge process. Results of electrochemical measurements, X-ray diffraction and transmission electron microscopy show that boron aids the formation of lithium intercalated MnO₂ (Li_xMnO₂) during discharge with some degree of reversibility, however, capacity fade and a steep voltage profile is observed. The addition of boron nitride (BN) improved the discharge performance and characteristics of the cell are also compared and discussed.

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1. Introduction

As fossil energy reserves are limited there is a need for secondary rechargeable batteries for energy storage to cater for the widespread energy requirements of the current global economy. Lithium-ion batteries are good candidates but they contain flammable organic electrolytes and are expensive. Therefore, the application of aqueous electrolyte was seen as an alternative approach, and much research has been focussed on developing aqueous batteries [1–5]. Recently, we showed that lithium intercalation can occur in aqueous media using lithium hydroxide as the electrolyte [6]. Among the candidates suitable as a cathode for these aqueous systems, one of the most attractive materials to replace cobalt oxide is manganese dioxide since the cost of manganese is low, non-toxic, has high voltage profile and exists in various polymorphic modifications [7–9]. Manganese oxides with these modifications (α , β , γ , and δ) have been widely studied as cathode materials for electrochemical energy storage systems but in non-aqueous organic solvents as electrolytes. Improvement on energy density is a challenge for application of MnO₂ material to be used in aqueous systems. The electrochemically active form of MnO₂ presently used in alkaline batteries is commonly referred to as γ -MnO₂.

As for alkaline battery using γ -MnO₂ as cathode with Zn as an anode, the improved electrochemical performance of γ -MnO₂

has opened the bottleneck to improve MnO₂ battery performance. Doping or physically mixing additives in battery electrodes is used normally to improve the electrochemical properties of a material during intercalation. It has been shown that introducing Bi³⁺ additive into the MnO₂ brings a slight improvement in the battery behavior [10]. Several other additives have also been explored [10,11]. As part of our continuing program to develop suitable cathode materials for an aqueous secondary battery system, we investigated a range of additives such as TiS₂, TiB₂, Bi₂O₃, CaO and CeO₂ [12–14]. These additives were shown to suppress the formation of manganese oxyhydroxides in the MnO₂ cathode and made the alkaline battery as a secondary system [12,13]. In our present contribution, we have chosen to physically add boron carbide (B₄C) or boron nitride (BN) to MnO₂ cathode for Zn–MnO₂ battery and test their electrochemical characteristics for long term cycling. Boron compounds are known to be excellent conductors and are generally doped in the carbon anode to accelerate the graphitization process which in turn enhances the performance of lithium batteries using non-aqueous electrolytes [15]. However, to the best of our knowledge, no work on boron as an additive has been pursued in aqueous electrolytes. Our current work is an extension to our recent preliminary study on boron carbide as an additive to MnO₂ in aqueous LiOH or KOH electrolytes [16].

2. Experimental

The EMD (electrolytic manganese dioxide; γ -MnO₂) type (IBA sample 32) material used in this work was purchased from the Kerr McGee Chemical Corporation. Boron carbide (B₄C) and boron nitride (BN) were obtained from Cerac Inc. For the electrochemical test, a pellet was prepared by mixing 65–75 wt.% MnO₂ with

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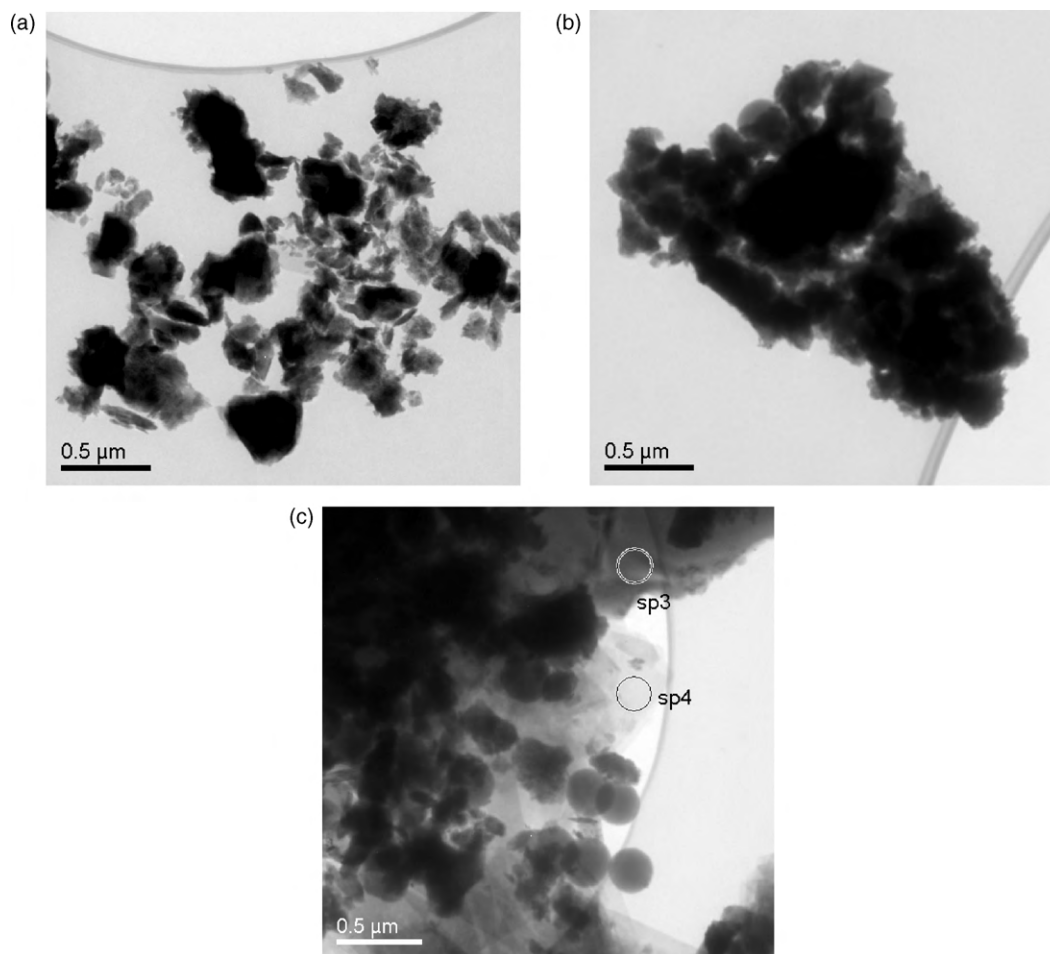


Fig. 1. TEM images of the B_4C modified MnO_2 cathode before discharge with different weight percent of additive content (a) 5 wt.%, (b) 7 wt.% and (c) 10 wt.%.

0–10 wt.% B_4C or BN with 15 wt.% acetylene black (A-99, Asbury, USA) and 10 wt.% poly (vinylidene difluoride) (PVDF, Sigma Aldrich) binder in a mortar and pestle. An electrochemical cell was constructed with the disk-like pellet as the cathode, Zn metal as the anode and filter paper as the separator. The electrolyte was a saturated solution of lithium hydroxide (LiOH) containing 1 mol l^{-1} zinc sulphate ($ZnSO_4$) with a pH equivalent to 10.5. The cell performance was evaluated galvanostatically at a current density of 0.5 mA cm^{-2} with the aid of 8 channel battery analyser from MTI Corporation, USA. The cell design and its experimental details were similar to those reported earlier [12].

The morphology of the products formed before and after discharge was characterized by transmission electron microscopy (TEM), energy dispersive X-ray spectroscopy analysis (EDS) using a JEOL 2010F TEM operated at 200 kV. TEM specimens were prepared by grinding a small fragment scraped from the pressed pellet under ethanol and dispersing on a holey carbon support film. A Siemens D500 X-ray diffractometer 5635 using $Co K\alpha$ radiation was used for X-ray diffraction analysis. The voltage and current were 40 kV and 30 mA, respectively. The scan rate was 1 degree per min between two theta values of 20° and 60° . Conductivity measurements for the B_4C modified MnO_2 samples were done on the four probe method.

3. Results and discussion

Bright field TEM images of boron carbide (B_4C) modified (5, 7 and 10 wt.%) MnO_2 particles before discharge are shown in Fig. 1. As expected, boron carbide is most visible in the TEM images for the highly loaded material 10 wt.% (Fig. 1c). The presence of B_4C in the location cited as “Sp3” in Fig. 1c was confirmed by EDS analysis whereas as the area labelled “Sp4” is composed of carbon. MnO_2 particles in the 5 wt.% B_4C added sample (Fig. 1a) appeared to be more discrete and the size distribution appeared finer compared with 7–10 wt.% added material. However, sampling difficulties associated with the TEM method make it difficult to be certain.

Fig. 2 compares the typical voltage response for the first galvanostatic cycle for B_4C modified MnO_2 containing 0, 5, 7 and 10 wt.% of additive content. All the cells are discharged under identical conditions. On commencement of discharge, the voltage dropped from the open-circuit voltage 1.85–1.75 V (for 0 wt.% additive) 1.7–1.69 V

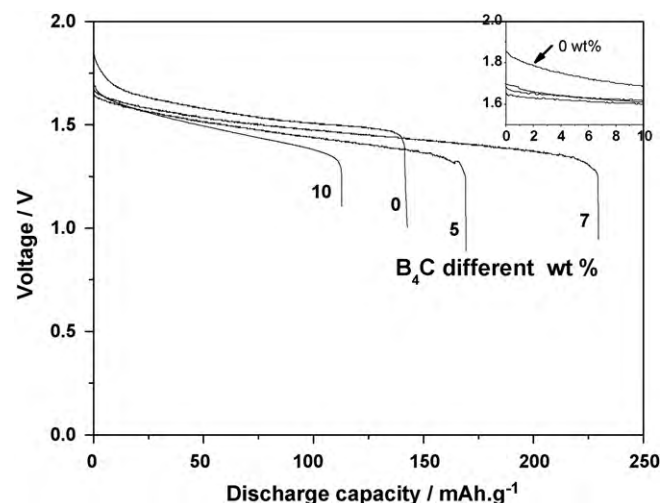


Fig. 2. Comparison of the first discharge profile of boron carbide modified MnO_2 cathode (different weight percent indicated in the profiles) using saturated aqueous LiOH containing 1 mol l^{-1} of $ZnSO_4$ as the electrolyte. Inset shows a difference in voltage profile for the additive free MnO_2 cathode.

Table 1
B₄C additive content, surface area and conductivity of prepared samples.

B ₄ C additive concentration/wt.%	Surface area/m ² g ⁻¹	Resistivity/(Ω cm)	Conductivity/(Ω cm) ⁻¹
0	40	8.571	0.1166
2	44	5.570	0.1795
5	51	5.304	0.1885
7	55	5.033	0.1986
10	57	4.681	0.2136

(for B₄C modified MnO₂) and then gradually decreased to 1.4 V and 1.3 respectively. The discharge was terminated when the voltage suddenly fell to 1.0 V. It is obviously seen in the inset (Fig. 2) that the voltage profile is higher for the additive free MnO₂, however the shape of the curves are same in both the 0 and 5 wt.% content samples. The significant capacity is highlighted in the presence of boron carbide additive modified to ≤ 7 wt.%. However, increasing the additive content above this amount to 10 wt.% causes a decrease in discharge capacity from 230 to 110 mAh g⁻¹. This shows the presence of boron carbide enhances the utilization of the MnO₂ material, as the rapid fall of potential occurs only at 1.3 V of plateau. In the absence of boron carbide the fall in potential occurs at a higher voltage of 1.4 V. These results could be due to one of the following possibilities: (1) boron is effective in enhancing the electrical conductivity of the active MnO₂ material that enhances the discharge capacity [17] or (2) since boron is a fast diffusant of light mass it can diffuse into the host MnO₂ and stabilize the MnO₆ octahedra [18]. This could strengthen the bond and increase the voltage

retention as seen in Fig. 2. As boron is physically mixed in the MnO₂ cathode without any heat treatment, it is unlikely that boron is substituted for MnO₂ in these samples.

It is also interesting to note that the data presented in Table 1 shows that surface area and conductivity of the B₄C modified MnO₂ are higher than that with 0 wt.% additive. It increases steeply for 2 wt.% and then it increased gradually for the remaining B₄C concentration. This supports that conductivity and higher surface area were accelerated by addition of B₄C in MnO₂ structure, resulting in enhancing discharge capacity from 140 to 230 mAh g⁻¹. However, too much B₄C ≥ 7 wt.% inhibits the entrance of the charge carriers to MnO₂ sites which results in decrease in discharge capacity. Therefore, only a small part of any pore is active and deeper into a pore (or penetration depth) results in a loss of capacity from 230 to 110 mAh g⁻¹ at higher B₄C concentration.

Hence, the nature of the MnO₂ cathode material after discharge was characterized using TEM, EDS and XRD techniques in order to elucidate the role of boron carbide. TEM images of the discharged

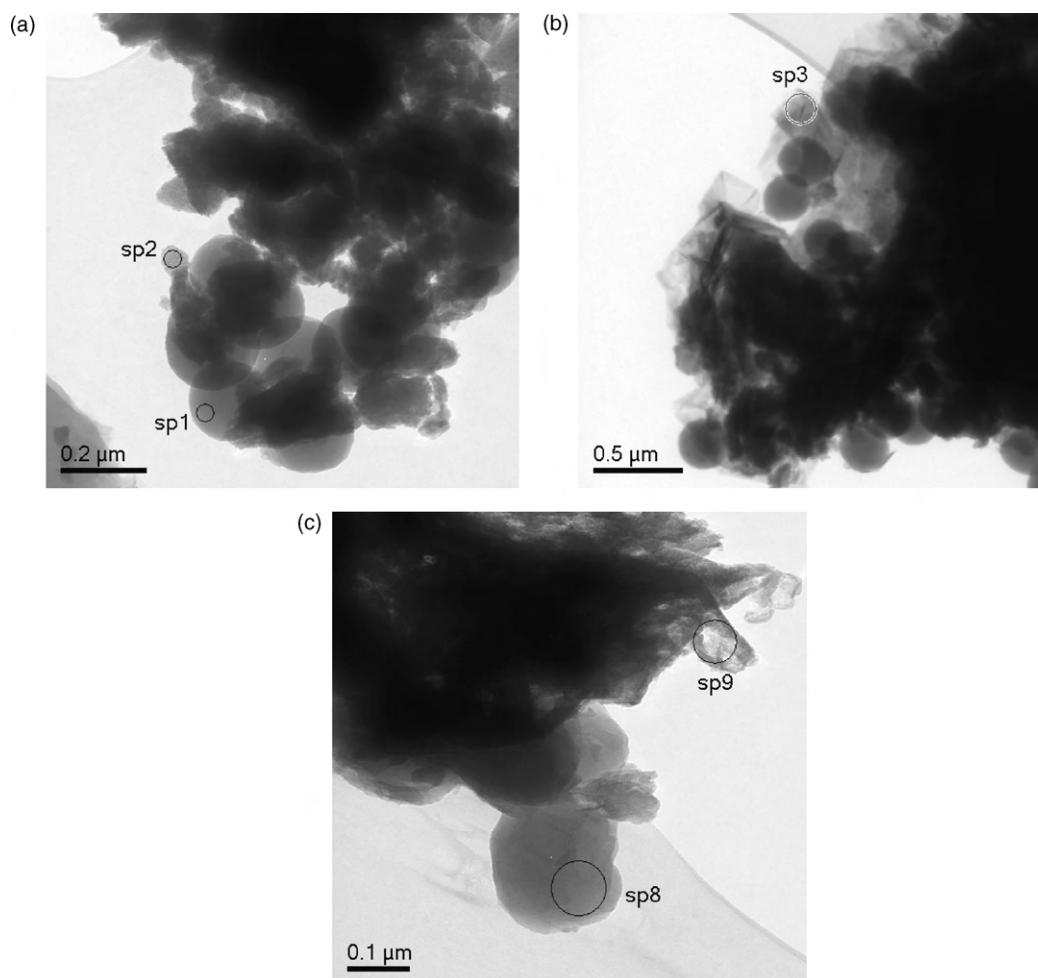


Fig. 3. TEM images of the discharged B₄C modified MnO₂ with different weight percent of the additive content (a) 5 wt.%, (b) 7 wt.% and (c) 10 wt.%.

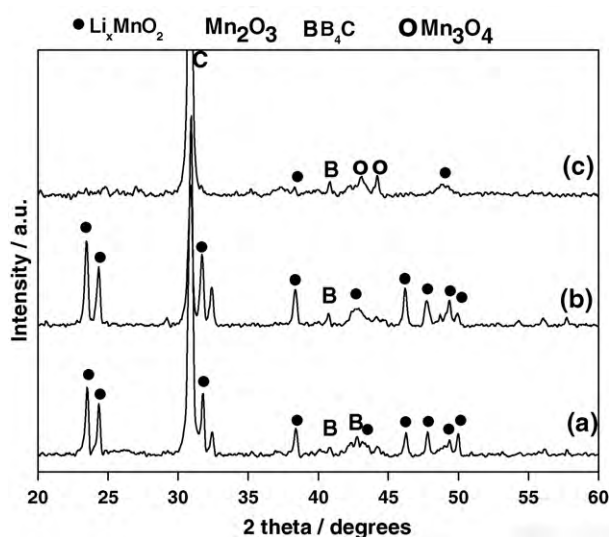


Fig. 4. XRD pattern of the discharged B_4C (5 wt.%) modified MnO_2 in LiOH electrolyte with different weight percent of the additive content (a) 5 wt.%, (b) 7 wt.% and (c) 10 wt.%.

B_4C modified MnO_2 cathodes are shown in Fig. 3. The bright field images (Fig. 3a and b) corresponding to 5 and 7 wt.% B_4C additives showed the MnO_2 to be crystalline while 10 wt.% B_4C additive showed a fibrous like pattern. The EDS analyses at the locations cited as “Sp9” (in Fig. 3) showed that the manganese present here are in different oxygen content to that of the locations cited as “Sp2 and Sp3”. This indicates the discharged product in 10 wt.% could be different from that of 5 and 7 wt.% additive.

To investigate the response of the MnO_2 structure during discharge, XRD patterns were recorded. The 5 and 7 wt.% cathode (Fig. 4a and b) showed the peaks are in good agreement with those reported for the following materials: Li_xMnO_2 (●) [19,20] and Mn_2O_3 (□) [21]. This shows that during discharge lithium is intercalated into the MnO_2 structure to form lithiated γ - MnO_2 [19,20] and also partly reduced to form manganese oxide (Mn_2O_3) as a minor phase. The main Bragg reflection corresponding to graphite is seen at $2\theta = 30^\circ$ (C). The presence of Li_xMnO_2 is further evidenced from Fig. 5 at the locations cited as “Sp4” and “Sp1”. The XRD pattern for a 10 wt.% additive content (Fig. 4c) showed new reflections corresponding to Mn_3O_4 (○) in addition to very few lithium intercalated MnO_2 (●) peaks. This indicates that too much boron carbide hinders

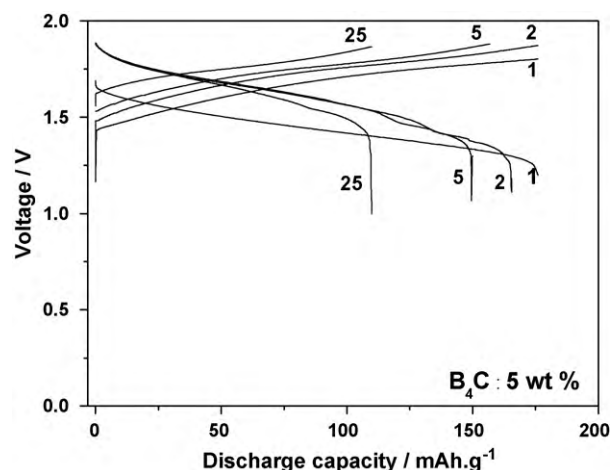


Fig. 6. Multiple discharge–charge behavior of B_4C modified MnO_2 cathode with 5 wt.% of additive content. The capacity of the cell decreases on cycling—cycle numbers shown.

the lithium intercalation into the MnO_2 structure and thus the discharge mechanism involves both lithium and proton intercalation [13]. The XRD results are in agreement with the electrochemical measurements and TEM results.

To investigate rechargeability, multiple galvanostatic discharge–charge cycles are performed on the B_4C modified (5, 7 and 10 wt.%) MnO_2 cathode. The results are shown in Figs. 6–8. All the cells were discharged and charged under identical conditions. The discharge–charge profiles for the first 25 cycles of the 5 wt.% additive content (in Fig. 6) showed a higher initial discharge capacity to that of the additive free (0 wt.%) MnO_2 cathode [12]. The cell discharge capacity initially dropped to 175 mAh g^{-1} and then to 165 mAh g^{-1} , after which it decreased gradually to 110 mAh g^{-1} following to 25th cycle. Nevertheless, the efficiency of the cell is still higher than the 0 wt.% MnO_2 [12]. The cycling performance for 7 wt.% additive content (in Fig. 7) showed a high irreversible performance during the consecutive cycles and the efficiency dropped to 46% in the 2nd cycle and quickly fades to 20% within the first 10 cycles. This implies the passivation of lithium species during cycling [22]. This could be explained in a way that 7 wt.% B_4C additive content accelerates more lithium intercalation but the process is not reversible. The cycling ability of 10 wt.% shown in Fig. 8 is very poor in behavior with respect

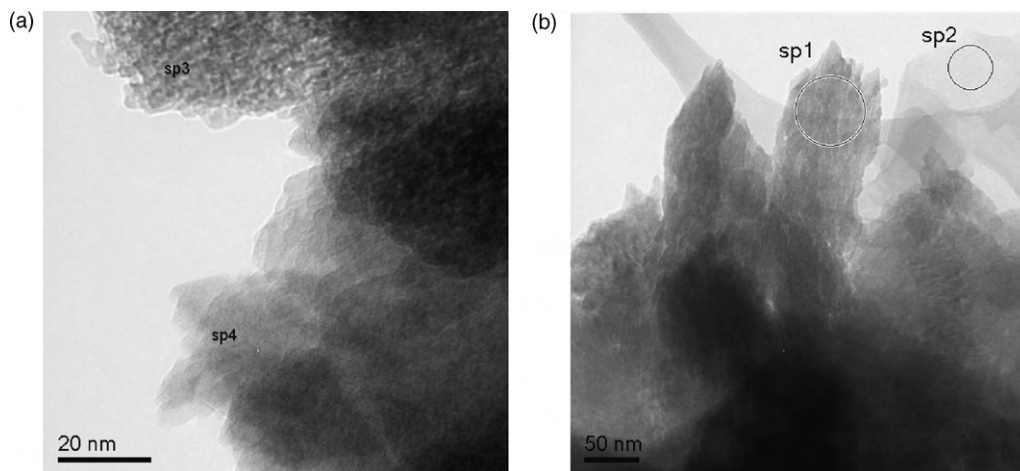


Fig. 5. TEM images of the discharged B_4C modified MnO_2 with the (a) 5 wt.% showing crystalline particles of Zn oxide (Sp3) and Mn oxide (Sp4) and (b) 7 wt.% showing manganese with the crystalline like patterns.

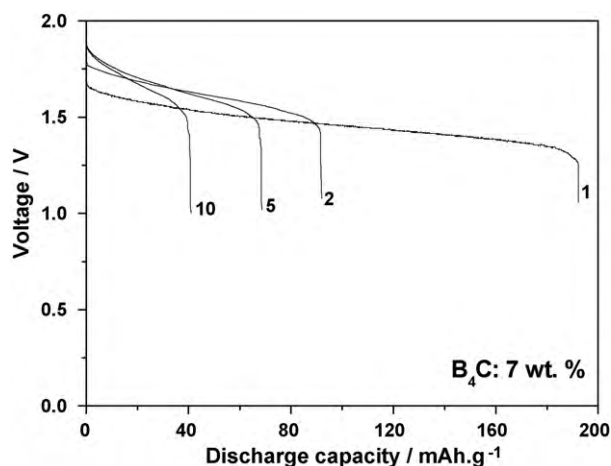


Fig. 7. Multiple discharge–charge behavior of B_4C modified MnO_2 cathode with 7 wt.% content. The capacity fades much more quickly on cycling—cycle numbers are shown.

to 0, 5 and 7 wt.% contents. This is simply due to excess amount of boron carbide blocks the entrance for the lithium ion to reach MnO_2 stable sites and the discharge mechanism proceeds with both lithium and proton insertion of MnO_2 .

3.1. Effect of boron nitride (BN) modified MnO_2

The effect of a boron nitride modified MnO_2 containing 5 wt.% of additive content is compared with the same amount of boron carbide (B_4C) in Fig. 9. The boron nitride modified MnO_2 electrode has enhanced the discharged capacity to 185 mAh g^{-1} with the mean discharge voltage of 1.5V, whereas the presence of boron carbide results in a lower capacity with a shift of 0.13 V to the lower discharge voltage. This could be understood in terms that during discharge i.e. upon lithium intercalation, boron carbide acts as electron acceptor and boron nitride acts as electron donor. This reflects in the energy of the lithium– MnO_2 bond which determines the intercalation cell potential. Hence the boron nitride electrode has the voltage increased compared with one with a boron carbide additive. Boron nitride strongly affects the reversibility of the MnO_2 cell (Fig. 10) with an abrupt decrease in cell discharge capacity from 185 to 25 mAh g^{-1} following to 10th cycling which is inferior to that of boron carbide counterpart. The irreversible capacity of boron

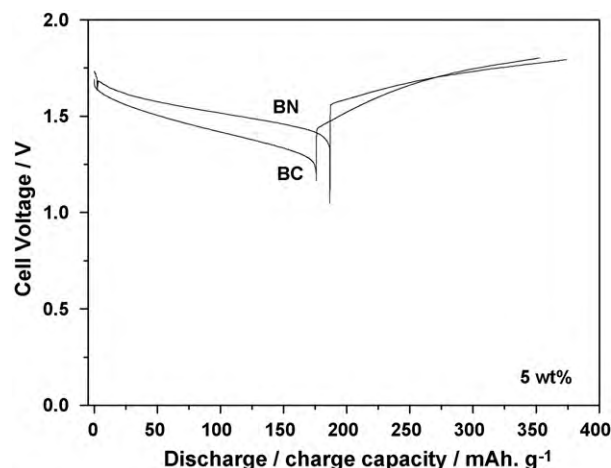


Fig. 9. Comparison of the first discharge–charge profiles of boron carbide or nitride modified (5 wt.%) MnO_2 cathode with zinc as an anode using saturated aqueous $LiOH$ containing 1 mol l^{-1} of $ZnSO_4$ as the electrolyte.

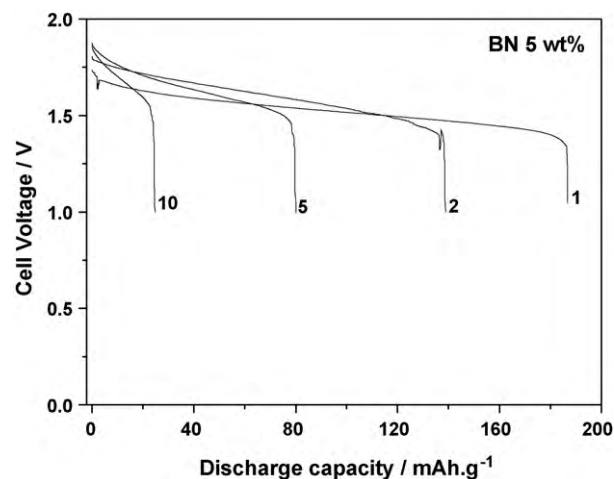


Fig. 10. Multiple discharge–charge behavior of BN modified MnO_2 cathode with 5 wt.% of additive content. The capacity fades quickly on cycling—cycle numbers are shown.

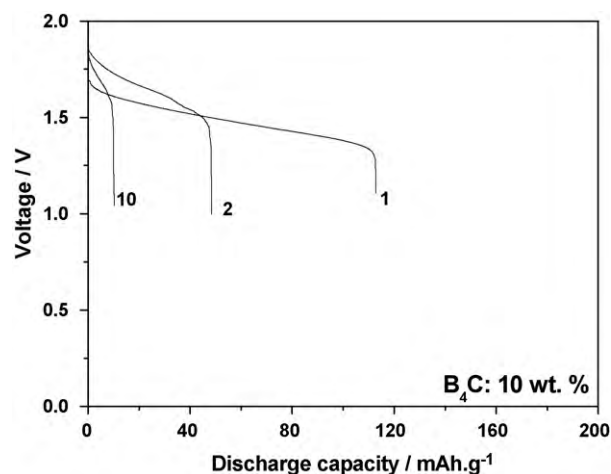


Fig. 8. Multiple discharge–charge behavior of B_4C modified MnO_2 cathode with 10 wt.% of additive content. The capacity fades very quickly on cycling—cycle numbers are shown.

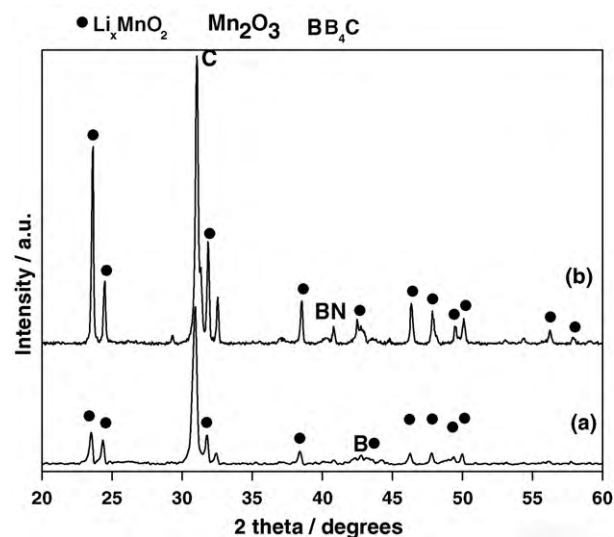


Fig. 11. XRD pattern of the discharged (a) B_4C and (b) BN modified MnO_2 cathode with 5 wt.% of additive content.

nitride electrode suggests that the lithium could react with nitrogen forming an unwanted product of lithium nitride (Li_3N) which is a well known reaction. The XRD pattern (in Fig. 11) of the boron nitride electrode supports the formation of lithium intercalated MnO_2 [20] however there is no evidence of Li_3N .

4. Conclusions

Boron carbide (B_4C) modified MnO_2 was used as cathode material in alkaline batteries and their electrochemical behavior was investigated. X-ray diffraction of the discharged product shows that boron carbide of 5–7 wt.% additives content produces lithium intercalated MnO_2 but ≥ 7 wt.% results in the formation of mixture of phases. The addition of 5 wt.% boron carbide to MnO_2 results in comparable rechargeability to that of 0 wt.% after 25 cycles, albeit with a much higher initial discharge capacity. Higher levels of additive content resulted in poorer cyclability. Boron nitride modified MnO_2 enhanced the discharge capacity and voltage, nevertheless this material is not suitable for secondary battery application. Hence, 5 wt.% of the B_4C modified MnO_2 could be of some value to the alkaline battery system.

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